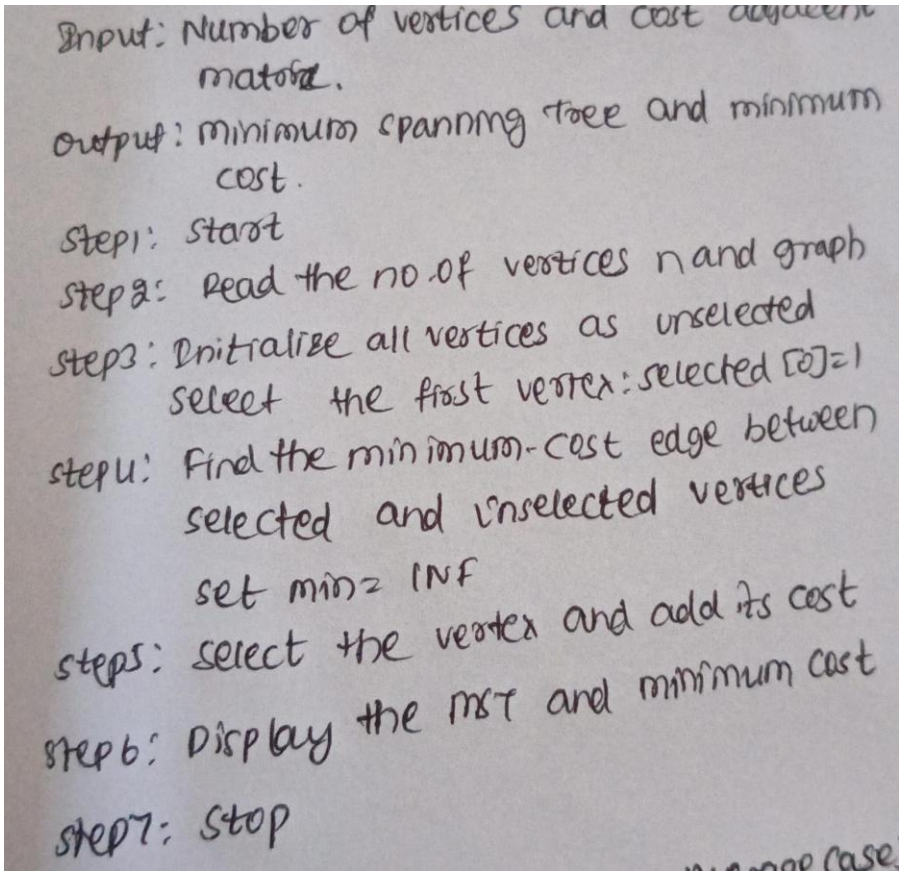
PRACTICAL-9

PRIM'S ALGORITHM

AIM: To Implement prim's algorithm

Algorithm:



PROGRAM:

```
#include <iostream>

using namespace std;


#define MAX 100

#define INF 9999


int main()
```

```
{
   int graph[MAX][MAX];
   int selected[MAX];
   int n, i, j;
   int edge = 0;
   int x = 0, y = 0;
   int min, cost = 0;

   cout << "Enter number of vertices: ";
   cin >> n;

   cout << "Enter Cost Adjacency Matrix:\n";

   for(i = 0; i < n; i++)
   {
      for(j = 0; j < n; j++)
      {
         cin >> graph[i][j];

         if(graph[i][j] == 0)
            graph[i][j] = INF;
      }
   }

   // Initially no vertex is selected
   for(i = 0; i < n; i++)
```

```
    selected[i] = 0;

// Select first vertex
selected[0] = 1;

cout << "\nEdges in Minimum Spanning Tree:\n";

while(edge < n - 1)
{
    min = INF;

    for(i = 0; i < n; i++)
    {
        if(selected[i])
        {
            for(j = 0; j < n; j++)
            {
                if(!selected[j] && graph[i][j] < min)
                {
                    min = graph[i][j];
                    x = i;
                    y = j;
                }
            }
        }
    }
```

```
    cout << x << " --> " << y
         << "  Cost = " << graph[x][y] << endl;

    cost += graph[x][y];

    selected[y] = 1;
    edge++;
  }

  cout << "\nMinimum Cost = " << cost << endl;

  return 0;
}
```

OUTPUT:

Output

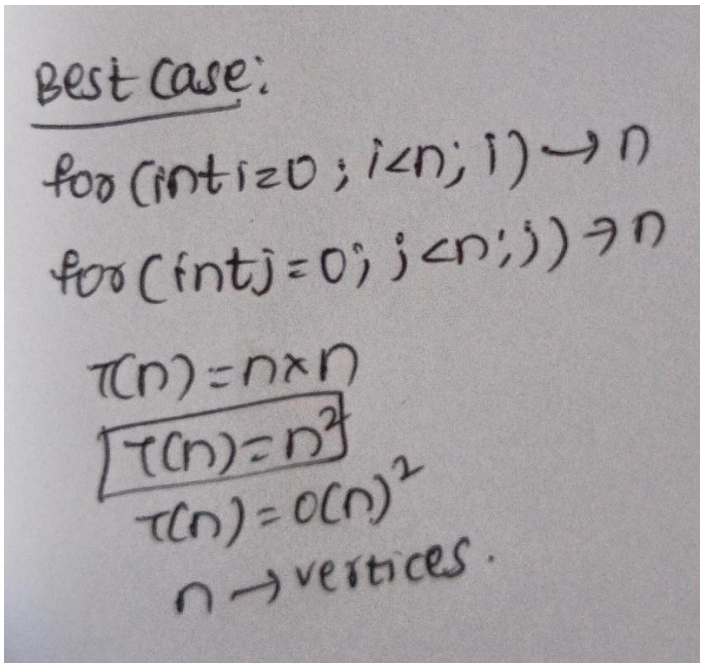
```
Enter number of vertices: 4
Enter Cost Adjacency Matrix:
0 2 3 0
2 0 1 4
3 1 0 5
0 4 5 0

Edges in Minimum Spanning Tree:
0 --> 1  Cost = 2
1 --> 2  Cost = 1
1 --> 3  Cost = 4

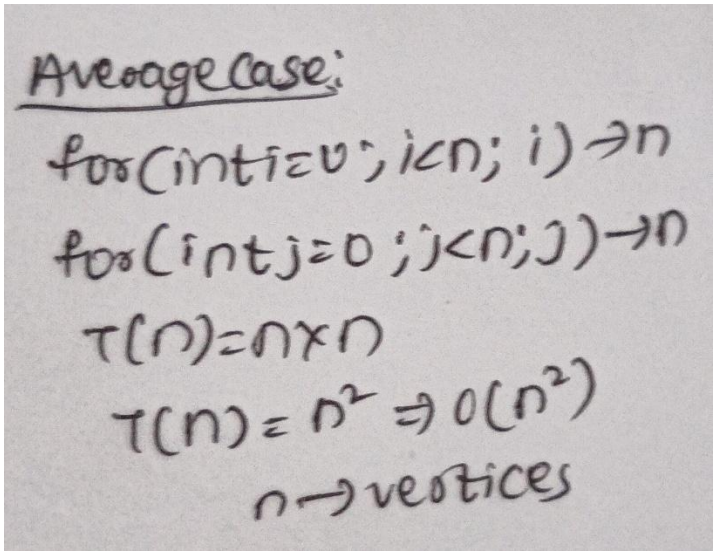
Minimum Cost = 7
```

TIME COMPLEXITY:

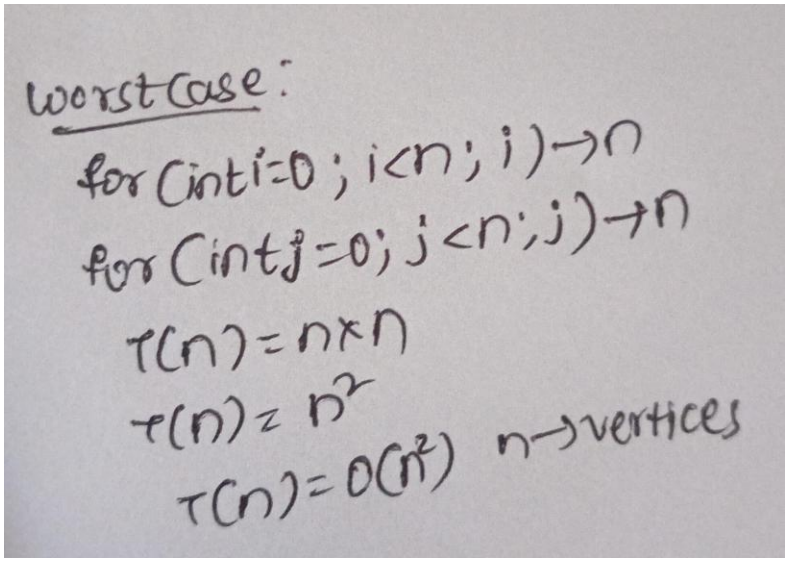
BEST CASE:



AVERAGE CASE:



WORST CASE:



CONCLUSION:

The implementation of Prim's Algorithm helps us understand how a Minimum Spanning Tree (MST) is constructed for a weighted graph. It repeatedly selects the minimum-cost edge connecting a selected vertex to an unselected vertex. For the given adjacency matrix implementation, Prim's Algorithm has a time complexity of $O(n^2)$ in the best, average, and worst cases. Therefore, Prim's Algorithm is a useful technique for finding the minimum-cost spanning tree of a connected graph.